Experiment 9

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Subject Name: System Design **Subject Code: 23CSH-314**

1. **Aim**: To design and analyze a scalable **Movie Recommendation System (like Netflix/Amazon Prime)** that suggests personalized movies to users based on preferences, watch history, and ratings using distributed system architecture.

2. **Objective**:

• To identify functional and non-functional requirements of a recommendation system
• To design High Level Architecture (HLD) for movie platforms
• To understand recommendation algorithms (content-based & collaborative filtering)
• To design microservices for user, movie, and recommendation management
• To study real-time user activity tracking
• To understand scalability challenges in large-scale systems

3. **Tools:**

Draw.io / Excalidraw -System design diagrams
MySQL / PostgreSQL - User & movie data
Redis - Caching
Kafka - Event streaming (user activity)
Load Balancer - Traffic distribution
Microservices Architecture
ElasticSearch - Fast searching
ML Models - Recommendation engine

4. **System Requirements:**

A. Functional Requirements:-

• Users should be able to register and login
• Users should view movie list and details
• Users should search movies
• Users should rate and review movies
• System should recommend personalized movies
• Users should create watchlist
• System should track watch history
• System should show trending/popular movies

B. Non-Functional Requirements:-

1) Scalability
a. Support millions of users
b. Handle large movie database

2) Consistency (CAP Theorem)
a. Eventual consistency acceptable
b. Accurate recommendations

3) Availability
a. Recommendations should always be available

4) Latency
a. Recommendation response < 200 ms
b. Search response < 100 ms

5) Reliability
a. No loss of user data
b. System should handle failures

5. High Level Design (HLD):

System follows Microservices Architecture

Main Components:

1) API Gateway
a. Authentication
b. Authorization
c. Routing
d. Rate Limiting

2) User Service
a. Registration & login
b. User profile management
c. Stores user preferences
d. User DB (MySQL)

3) Movie Service
a. Stores movie data
b. Fetch movie details
c. Movie DB

4) Recommendation Service
a. Provides personalized suggestions
b. Uses collaborative filtering
c. Uses content-based filtering

5) Rating & Review Service
   a. Stores ratings
   b. Stores reviews
   c. Updates recommendation system

6) Watchlist Service
   a. Add/remove movies
   b. Personalized list

7) Search Service
   a. Search by name, genre, actor
   b. Uses ElasticSearch

8) Activity Tracker (Kafka)
   a. Tracks user behavior
   b. Sends data to recommendation engine

9) Redis (Caching)
   a. Cache recommendations
   b. Cache popular movies

10) ML Model / Recommendation Engine
   a. Processes user data
   b. Generates recommendations

11) Database Layer
   a. User DB
   b. Movie DB
   c. Interaction DB

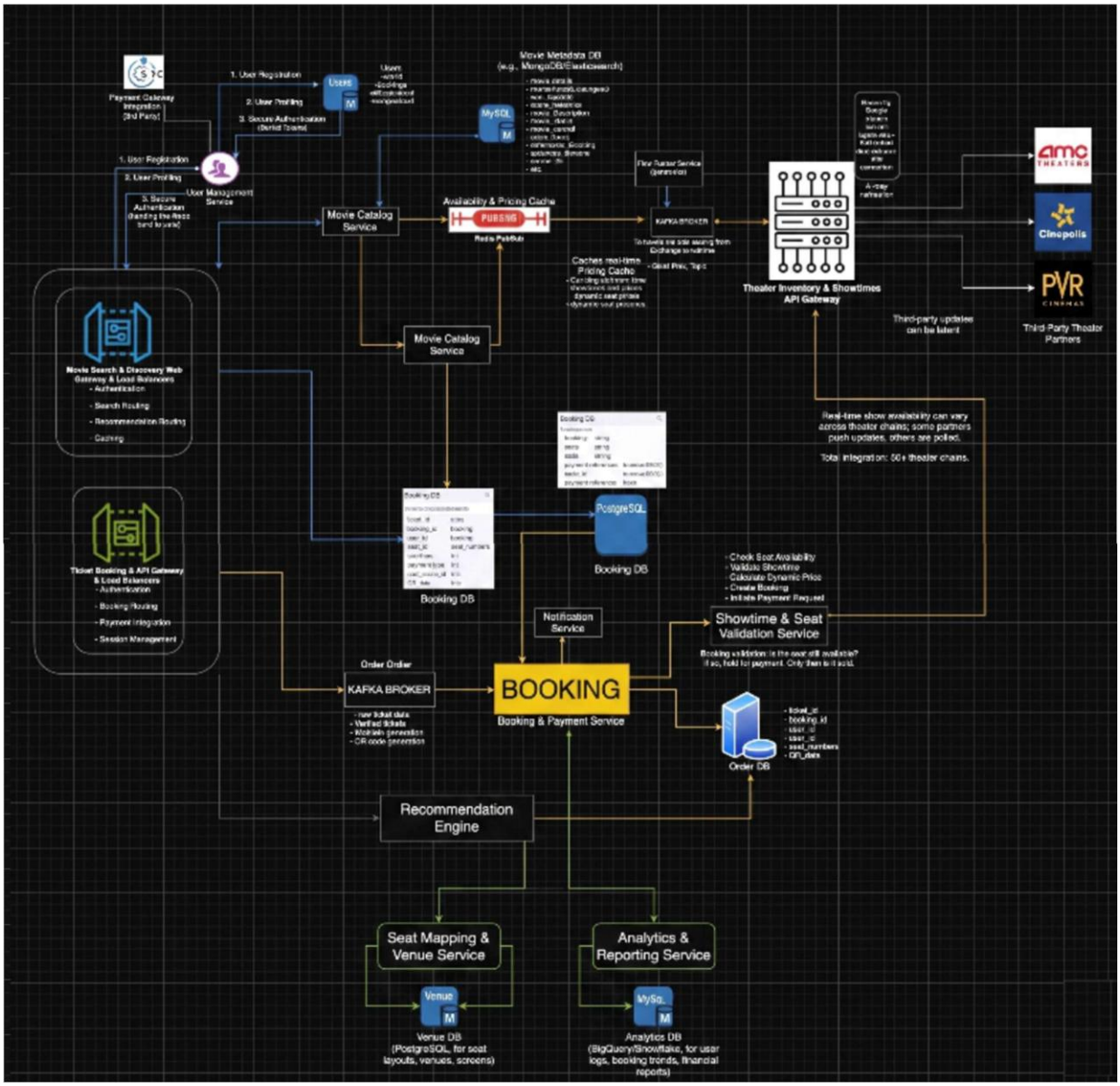
12) Load Balancer
   a. Distributes traffic
   b. Ensures high availability

6. Recommendation Flow (Step-by-Step):

- User logs in
- User browses or searches movies
- User watches/rates a movie
- Activity sent to Kafka
- Data processed by recommendation engine
- Personalized recommendations generated
- User receives recommended movies

7. Scalability Solution:

To support millions of users:
- Use Kafka for user activity streaming
- Use Redis for caching recommendations
- Use Load Balancers for traffic distribution
- Use Microservices architecture
- Use separate databases (User, Movie, Interaction)
- Use ElasticSearch for fast search
- Use API Gateway for centralized control



8. Learning Outcomes (What I Have Learnt):

- Learned how recommendation systems are designed
- Understood content-based and collaborative filtering
- Learned real-time user tracking using Kafka
- Understood personalized recommendation logic
- Learned importance of scalability in large systems
- Understood microservices architecture
- Learned how platforms like Netflix work